$\Rightarrow$ Surface $y^2 + z^2 = 2x^2$,

$\vec{p} = \hat{i} + \sqrt{2}\hat{k}$, $x = \sqrt{2}\sqrt{x^2+y^2}$,

$\Rightarrow$ $\vec{n}$, $-\vec{n}$, B

$\Rightarrow$ $2x^2 - y^2 - z^2 = 0$ not ~~FM~~

norm defn,

$v^T A v = 0$,

$\Rightarrow$ 'A is a diagonal matrix,

$\Rightarrow$ $v^T A v \rightarrow (2x - y - z)v$

$= 2x^2 - y^2 - z^2 = 0$.

$\Rightarrow$ A is $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

$\Rightarrow$ $v^T A v = 0$. $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

0,